

Task 1 - Normalisation

A Car hire firm, who are generally considered to be a shady company somehow earning larger profits, have hired you at a great expense to set up a new database system for them.

They have requested that you show then how you would use your knowledge of Normalization to design their new database system. The table below represents a snapshot of their customer car hire data. Each customer may hire cars from various outlets throughout Kerry. A car is registered at a particular outlet and can be hired out to a customer on a given date.

You should answer all questions that follow, below the table. You should also state clearly any assumptions you are making with respect to the data and your interpretation of the attributes listed.

carReg	make	model	custNo	custName	hireDate	outletNo	outletLoc
M565 OGD	Ford	Escort	C100	Smith, J	14/5/24	01	Killarney
M565 OGD	Ford	Escort	C201	Hen, P	15/5/24	01	Killarney
N734 TPR	Nissan	Sunny	C100	Smith, J	16/5/24	01	Killarney
M134 BRP	Ford	Escort	C313	Blatt, O	14/5/24	02	Tralelee
M134 BRP	Ford	Escort	C100	Smith, J	20/5/24	02	Tralelee
M611 OPQ	Nissan	Sunny	C295	Pen, T	20/5/24	02	Tralelee

- a) What stage of development is the above table currently in within the design process: UNF, 1NF, 2NF or BCNF? Explain your choice of answer.

The above table is currently in First Normal Form (FNF), each table cell has just one single value and the records are unique.

As there are no repeated values so it's not UNF (Unnormalized Form).

The table has partial dependencies for example, the make and model depend on carReg so it's not in Second Normal Form (2NF).

The table is not in Second Normal Form (2NF) so it can't be Boyce-Codd Normal Form (BCNF).

- b) The data in the above table is susceptible to processing anomalies. Provide and explain three examples each of how insertion, deletion and update anomalies could occur in the above table.

Insertion Anomalies occur when you can't add new data because some required information is missing

Insertion Anomalies might occur in the above table,

- If a new customer wants to hire a car but you can't enter their information as they haven't been assigned a car yet. This results in incomplete records.
- If a new outlet opens but has no cars yet, you might not be able to add the outlet's location because you also need the car details.
- If a customer wants to hire a car but doesn't have a customer number yet, you can't create their records until they get one. This could result in losing potential customer.

Deletion Anomalies occur when deleting a record from the database leads to the unintentional loss of other data.

Deletion Anomalies might occur in the above table,

- If you delete a car hire record, you might also accidentally delete all the customer's information if they have no other hire records. This means losing all their data.
- If you delete an outlet's record after all its hires are finished, you could lose the outlet's location info, even if it's still needed for future hires.
- If you were to delete a car hire record, it could remove historical data about how cars were used, making it harder to analyse trends in demand.

Update Anomalies occur when changing data leads to inconsistencies because multiple records needing to be updated.

Update Anomalies might occur in the above table,

- If an outlet's address changes, you need to update several records. If some are missed, you'll end up with different addresses for the same outlet.
- If you need to change details about a car model, and that car is mentioned in many hire records, not updating all of them can create confusion.
- If a customer changes their name and there are multiple records for them, you need to update all of them. If some are missed, it could lead to confusion about who the customer is.

- c) Using the process of Normalization, carry out the transformation of the above table to BCNF. All the steps should be clearly shown and labelled all keys, and functional dependencies should be identified at each stage. Identify the primary and foreign keys in your final BCNF relations.

Functional Dependencies – $\text{carReg} \rightarrow \text{make, model}$
 $\text{custNo} \rightarrow \text{custName}$
 $\text{hireDate, carReg, outletNo} \rightarrow \text{outletLoc}$
 $\text{outletNo} \rightarrow \text{outletLoc}$

Removed the partial dependencies

Created separate tables –

CarDetails Table – carReg, make, model

Customer Table – custNo, custName

CarHire Table – hireDate, carReg, custNo, outletNo

Ensure that every determinant is a candidate key

Created a separate table for outlet –

Outlet Table – outletNo, outletLoc

Boyce-Codd Normal Form (BCNF)

Table – CarDetails

Primary Key – carReg (Registration Number)

Attributes – carReg, make, model

Table – Customer

Primary Key – custNo (Customer Number)

Attributes – custNo, custName

Table – Outlet

Primary Key – outletNo (Outlet Number)

Attributes – outletNo, outletLoc

Table – CarHire

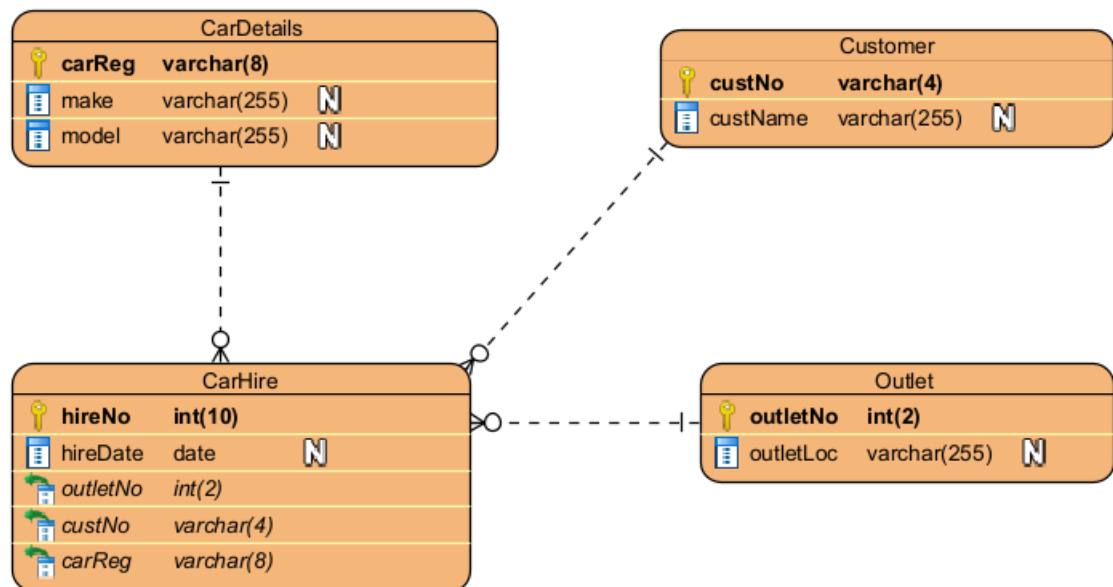
Primary Key – hireNo (Hire Number)

Foreign Keys – carReg (CarDetails Table),
custNo (Customer Table),
outletNo (Outlet Table)

Attribute – hireNo, carReg, custNo, outletNo, hireDate

Task 2 – Entity-Relationship model and MySQL database

- a) Draw an Entity-Relationship (E-R) model for the normalization process completed in step (c) above using the Visual paradigm software. Show all the entities, relationships, attributes and data in your design as per the outcomes from step (c). Save your completed visual paradigm project (.vpp) for inclusion in submission.



Data of CarDetails			Data of Customer		Data of Outlet	
carReg (PK)	make	model	custNo (PK)	custName		
M565 OGD	Ford	Escort	C100	Smith, J	outletNo (PK)	outletLoc
N734 TPR	Nissan	Sunny	C201	Hen, P	01	Killarney
M134 BRP	Ford	Escort	C313	Blatt, O	02	Tralee
M611 OPQ	Nissan	Sunny	C295	Pen, T		

Data of CarHire				
hireNo (PK)	hireDate	outletNo (FK)	custNo (FK)	carReg (FK)
58	14/05/2024	01	C100	M565 OGD
59	15/05/2024	01	C201	M565 OGD
60	16/05/2024	01	C100	M565 OGD
61	14/05/2024	02	C313	M134 BRP
62	20/05/2024	02	C100	M134 BRP
63	20/05/2024	02	C295	M611 OPQ

Instead of having one table with all the details of the Car Hire Company, I removed the partial dependencies and created four separate tables

- CarDetails Table – this table holds information about the cars available for hire, the car's make, model and registration number.
- Customer Table – this table contains information about the customers, their name and customer number.
- Outlet Table - this table holds information about the different outlets of the car hire company, the outlet number and location.
- CarHire Table – this table records when cars are hired. It includes the hire number, the date of hire, the outlet number, the customer number and the car registration number.

I created relationships between tables for example, the CarHire table connects to the Customer, Outlet and CarDetails tables.

I identified the Primary Keys for each table

- CarDetails Table Primary Key – regNo (Registration Number)
- Customer Table Primary Key – custNo (Customer Number)
- Outlet Table Primary Key – outletNo (Outlet Number)
- CarHire Table Primary Keys – hireNo (Hire Number)

Then I linked the tables using Foreign Keys in the CarHire Table

- CarHire Table Foreign Keys – regNo (CarDetails Table)
custNo (Customer Table)
outletNo (Outlet Table)

I used Visual Paradigm to create the E-R model using the 4 tables and keys.

Assumption Made

- Each car, customer and outlet are unique and has a unique identifier.
- The foreign keys in the CarHire table assume that,
A car must exist in the CarDetails table.
A customer must exist in the Customer table
An outlet must exist in the outlet table for every hire record.
- Each car can only be hired by one customer at a time, meaning a car can only be hired once on a given day.
- Every car hire transaction must be linked to an existing outlet to ensure each car has a physical location.
- It is assumed that the tables are filled with complete data to avoid any issues.

- b) Connect the E-R model in Visual paradigm with a database in MySQL. Populate the selected MySQL database with the above data. Export the data tables as MySQL database file (.sql) for inclusion in submission.

```
1 • SELECT * FROM assignement1.cardetails;
```

```
1 • SELECT * FROM assignement1.carhire;
```

Result Grid

carReg	make	model
M134 BRP	Ford	Escort
M565 OGD	Ford	Escort
M611 OPQ	Nissan	Sunny
N734 TPR	Nissan	Sunny

Result Grid

hireNo	hireDate	outletNo	custNo	carReg
58	2024-05-14	1	C100	M565 OGD
59	2024-05-15	1	C201	M565 OGD
60	2024-05-16	1	C100	M565 OGD
61	2024-05-14	2	C313	M134 BRP
62	2024-05-20	2	C100	M134 BRP

```
1 • SELECT * FROM assignement1.customer;
```

```
1 • SELECT * FROM assignement1.outlet;
```

Result Grid

custNo	custName
C100	Smith, J
C201	Hen, P
C295	Pen, T
C313	Blatt, O

Result Grid

outletNo	outletLoc
1	Killarney
2	Tralee

Connected the E-R diagram on visual paradigm to MySQL database under a schema called 'assignement1.' Added the data from the original table into the correct tables in MySQL.

Task 3 – Querying and Data manipulation

Write the following SQL queries to accomplish the provided tasks. Execute them on your MySQL database and include both the SQL code (in .txt format) and the query results as screenshots or text output in the report) for your submission.

- a) Identify the customer number to retrieve information.

Write a query to retrieve all cars hired by a specific customer, giving their custNo.

The query should return the customer's name, car registration, make, model, hire date and outlet location.

```
1 • USE assignment1;
2
3 /* Task 3 (a) */
4
5 • SELECT customer.custName, carhire.carReg, cardetails.make, cardetails.model, carhire.hireDate, outlet.outletLoc
6 FROM carhire
7 JOIN customer ON carhire.custNo = customer.custNo
8 JOIN cardetails ON carhire.carReg = cardetails.carReg
9 JOIN outlet ON carhire.outletNo = outlet.outletNo
10 WHERE customer.custNo = 'C100';
11
```

custName	carReg	make	model	hireDate	outletLoc
Smith, J	M565 OGD	Ford	Escort	2024-05-16	Killarney
Smith, J	M565 OGD	Ford	Escort	2024-05-14	Killarney
Smith, J	M134 BRP	Ford	Escort	2024-05-20	Tralee

- b) Calculate total hires for each customer

Write a query that calculates how many times each customer has hired a car. Return the customer number, customer name and the total number of hires.

```
1 • USE assignment1;
2
3 /* Task 3 (b) */
4
5 • SELECT carhire.custNo, customer.custname, COUNT(*) AS totalHires
6 FROM carhire
7 JOIN customer ON carhire.custNo = customer.custNo
8 GROUP BY custNo, custName;
9
```

custNo	custname	totalHires
C100	Smith, J	3
C201	Hen, P	1
C295	Pen, T	1
C313	Blatt, O	1

c) Update car model information

One of the car models existing in database needs to be updated. Write a query to update the model of a specific car registration (carReg = 'M565 0GD') from 'Escort' to 'Fiesta'

```
1 • USE assignement1;
2
3   /* Task 3 (c) */
4
5 • UPDATE cardetails
6   SET model = 'Fiesta'
7   WHERE carReg = 'M565 0GD'
```

carReg	make	model
M134 BRP	Ford	Escort
M565 0GD	Ford	Fiesta
M611 0PQ	Nissan	Sunny
N734 TPR	Nissan	Sunny

d) Delete rental records older than a specific date

Write a query to delete all records of car hires that occurred before 16th May 2024

```
1 • USE assignement1;
2
3   /* Task 3 (d) */
4
5 • DELETE FROM carhire
6   WHERE hireDate < '2024/05/16'
```